

BONUS: REVENUE OPTIMIZATION FOR VENUE V_GAMMA (The Snob Pit)

LOGICAL RESULT we should expect from our below calculations at the Venue V_Gamma.

We know from EDA that crowd energy has very weak correlation with ticket price i.e. they are not affected by the high prices of tickets.

And also on average there is higher crowd energy at V_Gamma for higher ticket prices.

And also crowd size and crowd energy have weak correlation.

So, logically we should expect optimal price of ticket to be quite high and the associated crowd size will also be at the higher side.

Assumptions:

We assumed relation between crowd energy and crowd size to be linear (as the plot of crowd energy v/s crowd size in EDA was quite complex).

So, we just created a linear model: Attendance (say x) = $\min(800, C * \text{crowd_energy})$, where C is mean crowd size/mean crowd energy.

So, Profit will be $xp - (5000 + 8x)$ where p is the ticket price.

Since we had to calculate the energy without inputting the crowd size (attendance) we just used median to nullify the effect otherwise it would have been a circular logic.

```
[62]: c = train["Crowd_Size"].mean() / train["Crowd_Energy"].mean()
      print("C:", c)

      scenario = (
          X[X["Venue_ID_V_Gamma"] == 1]
          .median()
          .to_frame()
          .T
      )
      import numpy as np

      prices = np.linspace(10, 120, 100)

      profits = []
      attendances = []
      energies = []

      for p in prices:
          scenario["Ticket_Price"] = p

          # Predict crowd energy using our model
          predicted_energy = best_model.predict(scenario)[0]

          # Convert energy to attendance
```

```

attendance = min(800, c * predicted_energy)

# Profit calculation
profit = attendance * p - (5000 + 8 * attendance)

energies.append(predicted_energy)
attendances.append(attendance)
profits.append(profit)

profits = np.array(profits)
best_idx = np.argmax(profits)

optimal_price = prices[best_idx]
max_profit = profits[best_idx]
optimal_attendance = attendances[best_idx]
optimal_energy = energies[best_idx]

print(f"Optimal Ticket Price: {optimal_price:.2f}")
print(f"Max Profit: {max_profit:,.0f}")
print(f"Expected Attendance: {optimal_attendance:.0f}")
print(f"Predicted Crowd Energy: {optimal_energy:.1f}")

```

```

C: 9.515129545809092
Optimal Ticket Price: 120.00
Max Profit: 75,768
Expected Attendance: 721
Predicted Crowd Energy: 75.8

```

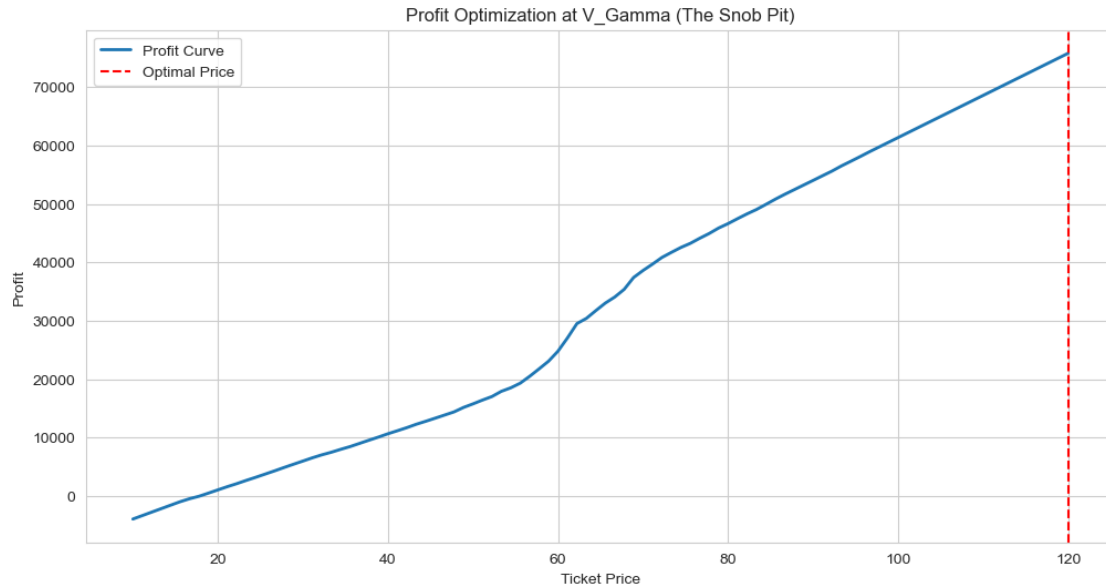
```

[63]: import matplotlib.pyplot as plt

plt.figure(figsize=(12,6))
plt.plot(prices, profits, linewidth=2, label="Profit Curve")
plt.axvline(optimal_price, color="red", linestyle="--", label="Optimal Price")

plt.xlabel("Ticket Price")
plt.ylabel("Profit")
plt.title("Profit Optimization at V_Gamma (The Snob Pit)")
plt.legend()
plt.grid(True)
plt.show()

```



NOW LET USE PREDICT OPTIMAL TICKET PRICE DIRECTLY USING TRAINING DATASET NOT OUR MODEL

```
[64]: vg = train[
    (train["Venue_ID"] == "V_Gamma") &
    (train["Crowd_Size"] <= 800) &
    (train["Ticket_Price"] >= 0) &
    (train["Ticket_Price"] <= 120)
].copy()
```

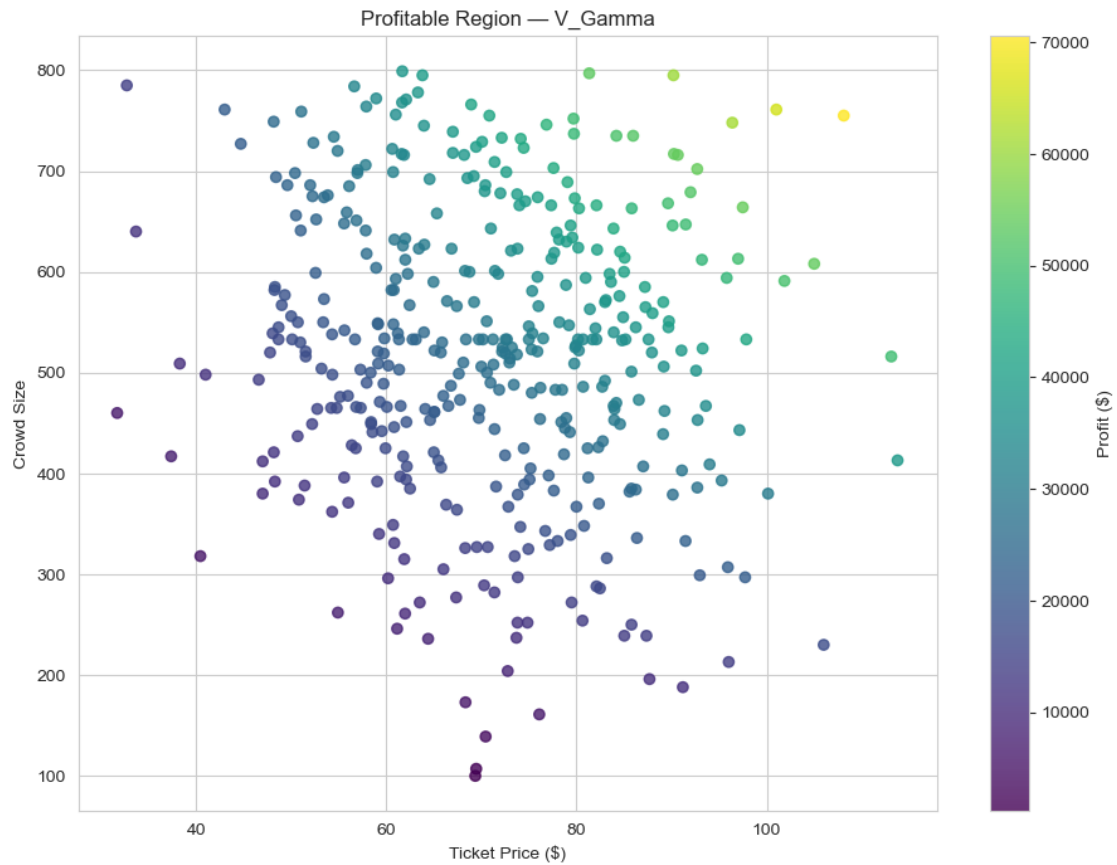
```
[65]: vg["Profit"] = (
    vg["Crowd_Size"] * vg["Ticket_Price"]
    - (5000 + 8 * vg["Crowd_Size"])
)
```

```
[66]: profitable = vg[vg["Profit"] > 0]

plt.figure(figsize=(11, 8))
plt.scatter(
    profitable["Ticket_Price"],
    profitable["Crowd_Size"],
    c=profitable["Profit"],
    cmap="viridis",
    alpha=0.8
)

plt.colorbar(label="Profit ($)")
plt.xlabel("Ticket Price ($)")
```

```
plt.ylabel("Crowd Size")
plt.title("Profitable Region - V_Gamma")
plt.show()
```



The above graph also agrees with our mathematical model which we created for optimal ticket pricing.

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